

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server stores and manages important customer and business information used by employees for daily operations. Because the server is publicly accessible, unauthorized users may attempt to access, modify, or disrupt the data it contains. Securing the server is important to protect sensitive information and maintain reliable business operations. If the server becomes unavailable or compromised, employees may be unable to perform essential tasks, and the company could experience financial and reputational damage.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information through unauthorized access or data exfiltration	3	3	9
Employee	Disrupt critical database operations through misuse or compromised credentials	2	3	6
Unauthorized user/customer	Modify or delete important database information	1	3	3

Approach

The assessment focused on risks related to the storage, access, and availability of information in the database. Threat sources and events were selected based on the server's public exposure and the likelihood that unauthorized users could attempt to access or disrupt it. The potential impact of each threat was evaluated against the confidentiality, integrity, and availability of business data. Higher-risk events were those most likely to cause significant operational, financial, or reputational damage.

Remediation Strategy

The organization should implement authentication, authorization, and auditing controls to ensure that only approved users can access the database. Strong passwords, multi-factor authentication, role-based access control, and least-privilege permissions should be enforced. Public access should be restricted through firewall rules, IP allow-listing, secure remote access, and network segmentation. Data in transit should use current TLS encryption, and regular logging, monitoring, patching, and backups should be maintained.